

Section: Our Dynamic Universe

Topic: The Expanding Universe

Astronomers observe light from a distant galaxy and find that its spectral lines are shifted towards the red end of the spectrum.

(a) Name this phenomenon.

✓ This is called a **redshift**.

(b) Explain what this redshift indicates about the motion of the galaxy relative to Earth.

✓ A redshift means the **wavelengths of the light have increased**.

This indicates the galaxy is **moving away** from the Earth — the light is stretched as it travels.

(c) The light from the galaxy has a redshift $z = 0.015$.

Calculate the speed of the galaxy relative to Earth.

Use the formula:

$$z = \frac{v}{c} \Rightarrow v = zc = 0.015 \times 3.00 \times 10^8 = 4.5 \times 10^6 \text{ m s}^{-1}$$

(d) State what is meant by the term “Hubble’s Law”.

✓ Hubble’s Law states that:

The **recessional velocity** of a galaxy is **directly proportional to its distance** from Earth.

Mathematically:

$$v = H_0 d$$

where H_0 is the Hubble constant.

🧠 **Revision Tips**

- A redshift means an object is moving **away**; blueshift means it’s **approaching**.
- Use $z = \frac{v}{c}$ for small redshifts (non-relativistic speeds).
- Hubble’s Law explains the **expansion of the universe**.
- The higher the redshift, the **faster** the galaxy is receding — and usually, the **farther away** it is.